

Q.1 During elongation, the enzyme adds _____ nucleotide at a time to the growing RNA transcript

- A. 2
- B. 3
- C. 1**
- D. 5

Q.2 The “transient excursion” model proposes transient cycles of forward and reverse translocation of _____

- A. RNA polymerase**
- B. DNA polymerase

Q.3 It is the addition of an initiation factor called σ that converts core enzyme ($\alpha 2 \beta \beta'$) into the form that initiates only at promoters. This form of the enzyme is called.....

- A. RNA polymerase holoenzyme**
- B. DNA polymerase holoenzyme
- C. DNA topoisomerase holoenzyme
- D. RNA topoisomerase holoenzyme

Q.4 RNA polymerase starts most transcripts with _____ ribonucleotide

- A. thymine
- B. adenine**

Q.5 TFIIB protein is a _____ polypeptide chain, enters the preinitiation complex

- A. double
- B. single**

Q.6 TAF42 and TAF62 from *Drosophila* have been shown to form a structure similar to that of the _____ tetramer of histones

- A. H3.H4**
- B. H1.H4
- C. H3.H2
- D. H1.H3

Q.7 Bacteria have _____ types of RNA polymerase

- A. 3
- B. 1**
- C. 5
- D. 2

Q.8 _____polymerase can initiate a new RNA chain on a DNA template

A. RNA

B. DNA

Q.9 During initial transcription, RNA polymerase produces and releases short RNA transcripts of _____

A. <10 nucleotides

B. <5 nucleotides
